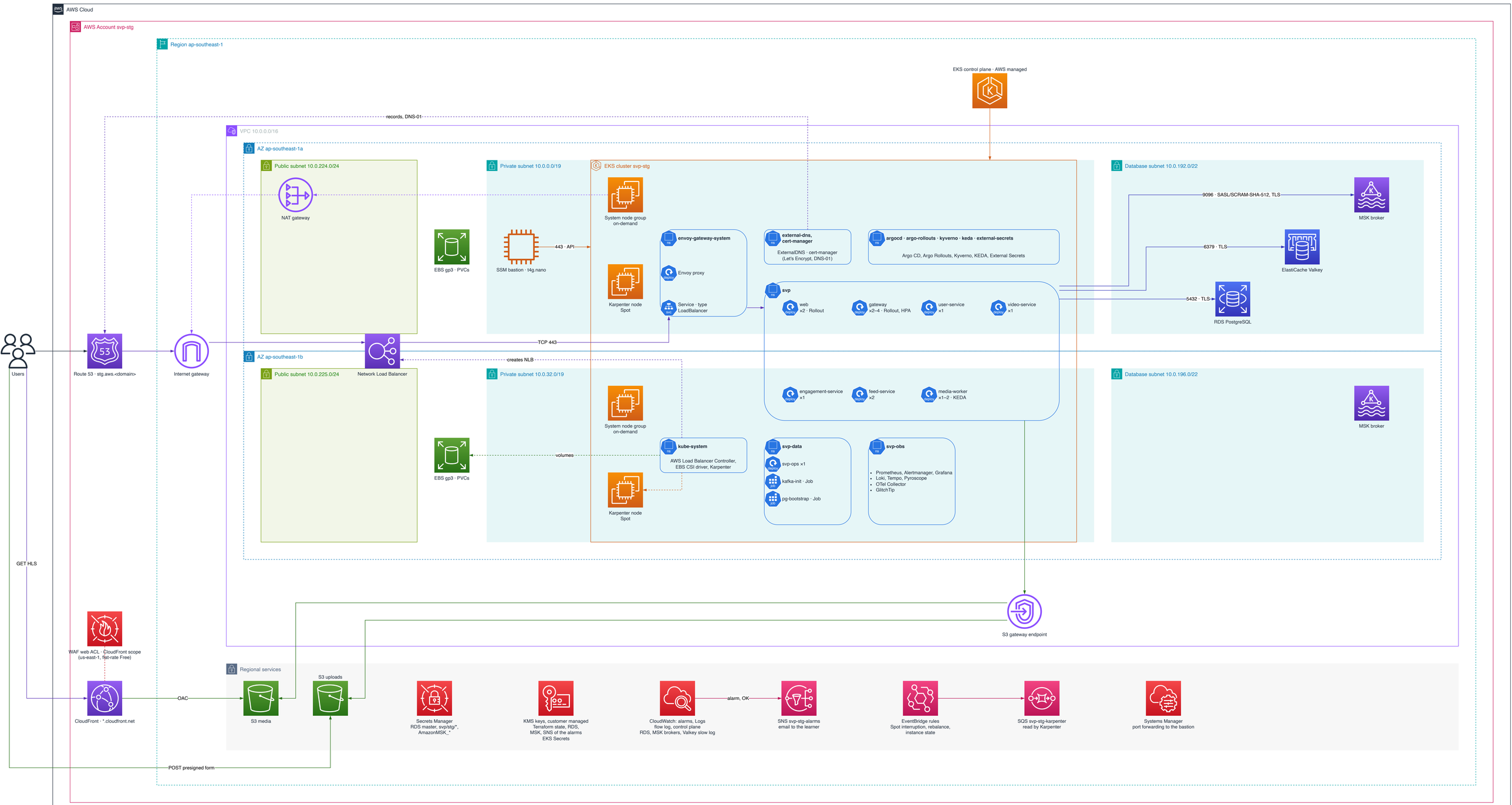
stg — EKS cluster svp-stg

ap-southeast-1 · 2 AZs, each with a public, a private and a database subnet · one NAT gateway, in the first AZ, for every private subnet · system node group on-demand and Karpenter's nodes (Spot), one icon each per AZ · API endpoint private only, reached through Systems Manager port forwarding to the SSM bastion · one NLB across all public subnets, made by the AWS Load Balancer Controller · EBS volumes in the AZ of their pod's node · TLS certificates from Let's Encrypt (cert-manager), no ACM

Database subnets, no route to the Internet: RDS single-AZ · Amazon MSK one broker per AZ, SASL/SCRAM-SHA-512 over TLS · ElastiCache for Valkey one node · AWS picks the AZs of RDS and ElastiCache (drawn from the first) · CloudWatch alarms of RDS, MSK and ElastiCache notify an SNS topic encrypted with a customer managed key, which emails the learner · KMS, Secrets Manager, CloudWatch, SNS, SQS, EventBridge regional

Kubernetes: a frame per namespace, an icon per workload with its replicas · the scheduler puts each pod on any node of any AZ, so the row of a frame or an icon says nothing about where its pods run (svp is drawn across the first two AZs to show it) · third-party tools as text · EKS add-ons (CoreDNS, kube-proxy, VPC CNI) not drawn

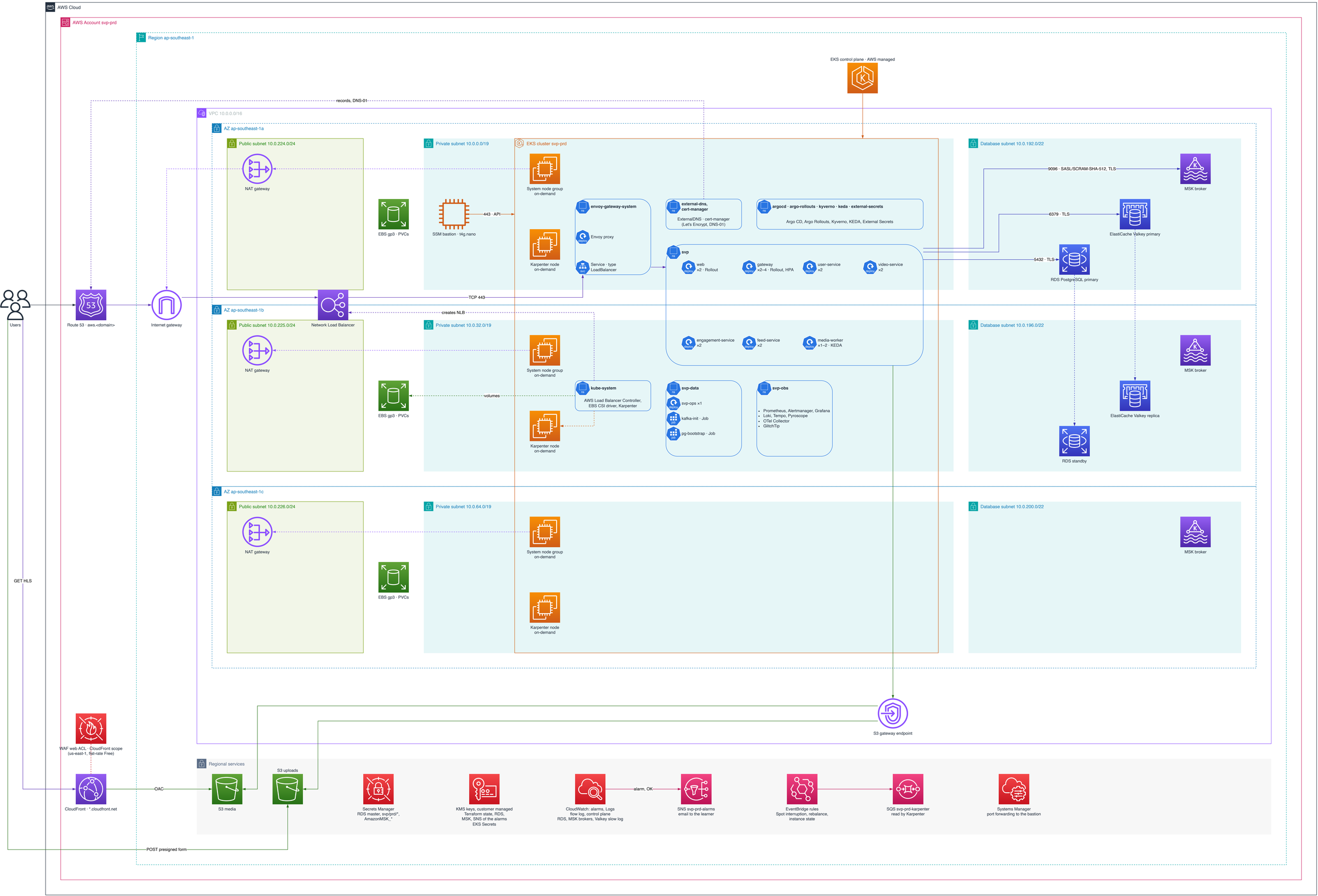


prd — EKS cluster svp-prd

ap-southeast-1 · 3 AZs, each with a public, a private and a database subnet · one NAT gateway per AZ · system node group on-demand and Karpenter's nodes (on-demand), one icon each per AZ · API endpoint private only, reached through Systems Manager port forwarding to the SSM bastion · one NLB across all public subnets, made by the AWS Load Balancer Controller · EBS volumes in the AZ of their pod's node · TLS certificates from Let's Encrypt (cert-manager), no ACM

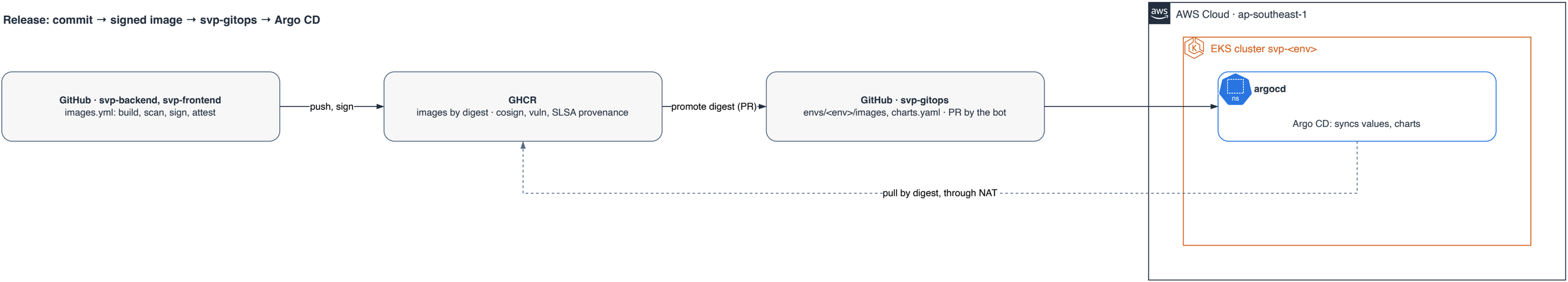
Database subnets, no route to the Internet: RDS Multi-AZ primary/standby · Amazon MSK one broker per AZ, SASL/SCRAM-SHA-512 over TLS · ElastiCache for Valkey primary and 1 replica(s), Multi-AZ, automatic failover · AWS picks the AZs of RDS and ElastiCache (drawn from the first) · CloudWatch alarms of RDS, MSK and ElastiCache notify an SNS topic encrypted with a customer managed key, which emails the learner · KMS, Secrets Manager, CloudWatch, SNS, SQS, EventBridge regional

Kubernetes: a frame per namespace, an icon per workload with its replicas · the scheduler puts each pod on any node of any AZ, so the row of a frame or an icon says nothing about where its pods run (svp is drawn across the first two AZs to show it) · third-party tools as text · EKS add-ons (CoreDNS, kube-proxy, VPC CNI) not drawn



Flows — release path and upload to playback

Release: commit → signed image → svp-gitops → Argo CD



Upload to playback

